ELEC40012 - MATHEMATICS1A - SOLUTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex numbers $z_1 = e^{i\pi/6}$ and $z_2 = e^{-i\pi/3}$, in cartesian form,

   (a) $z_1 - z_2 = \dfrac{\sqrt{3}}{2} - \dfrac{1}{2} - \left(\dfrac{\sqrt{3}}{2} + \dfrac{1}{2}\right) i$, (b) $z_1 z_2 = \dfrac{1}{2} - \dfrac{\sqrt{3}}{2} i$,

   (c) $z_1 + z_2 = \dfrac{\sqrt{3}}{2} + \dfrac{1}{2} + \left(\dfrac{\sqrt{3}}{2} - \dfrac{1}{2}\right) i$, (d) $z_2/z_1 = -i$,

   (e) none of the above [3]

   Solution Use Euler's formula
   $z_1 = \cos(\pi/6) + i\sin(\pi/6) = \dfrac{\sqrt{3}}{2} + \dfrac{1}{2}i$ and $z_2 = \cos(-\pi/3) + i\sin(-\pi/3) = \dfrac{1}{2} - \dfrac{\sqrt{3}}{2}i$,
   so neither (a) nor (c). $z_1 z_2 = e^{i\pi/6} e^{-i\pi/3} = e^{i(\pi/6 - \pi/3)} = e^{-i\pi/6} = \dfrac{\sqrt{3}}{2} - \dfrac{1}{2}i$ and $z_2/z_1 = e^{i(-\pi/3 - \pi/6)} = e^{-i\pi/2} = -i$ so (d)

2. Which of the following is the locus $z = x + iy$, in the complex plane, satisfying

   $$|z - i| = 2|i(z+1)| \ ?$$

   (a) A parabola with equation $x = -\dfrac{2}{3}y^2 + \dfrac{1}{9}y - \dfrac{4}{3}$;

   (b) a circle with centre $\left(-\dfrac{4}{3}, -\dfrac{1}{3}\right)$ and radius $\dfrac{2}{3}\sqrt{2}$;

   (c) a circle with centre $\left(\dfrac{1}{3}, \dfrac{4}{3}\right)$ and radius $\dfrac{1}{3}\sqrt{2}$;

   (d) an ellipse with equation $2(x-2)^2 + 3(y-1)^2 = 1$;

   (e) none of the above [3]

   solution Rewrite as

$$|x + i(y-1)| = 2|-y + i(x+1)| \Rightarrow x^2 + y^2 - 2y + 1 = 4(y^2 + x^2 + 2x + 1) \Rightarrow x^2 + \frac{8}{3}x + y^2 + \frac{2}{3}y = -1$$

   and complete the squares on the right-hand side to get $(x + 4/3)^2 + (y + 1/3)^2 = 8/9$, so (b)

3. Which of the following statements about $\lim_{x\to\infty} x\left(\sqrt{x^2+4}-x\right)$ is true?

(a) The limit is equal to 0 ; (b) the limit does not exist ;

(c) the limit is equal to 2 ; (d) the limit cannot be determined .

(e) none of the above [3]

solution Rewrite as

$$x\left(\sqrt{x^2+4}-x\right) = x\left(\sqrt{x^2\left(1+\frac{4}{x^2}\right)}-x\right) = x^2\left(\left[1+\frac{4}{x^2}\right]^{\frac{1}{2}}-1\right)$$

and use binomial expansion:

$$= x^2\left(1+\frac{1}{2}\left[\frac{4}{x^2}\right]+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left[\frac{4}{x^2}\right]^2+\cdots-1\right) = x^2\left(\frac{1}{2}\left[\frac{4}{x^2}\right]+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left[\frac{4}{x^2}\right]^2+\cdots\right)$$

$$= \frac{1}{2}(4)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left[\frac{16}{x^2}\right]+\cdots$$

and taking the limit as $x\to\infty$, we are left with 2, so (c).

4. The limit $\lim_{x\to 0}\dfrac{x^2}{\ln(\cos x)}$

(a) exists, but cannot be determined, ; (b) the limit does not exist ;

(c) is equal to -2 ; (d) is equal to 0 .

(e) none of the above [3]

solution: (c). Use l'Hopital's rule, twice.

5. The graph of the function $y = x^5 - 5x^4$ has

(a) a point of inflection at $x = 0$, and a local minimum at $x = 4$,

(b) points of inflection at $x = 0$ and $x = 3$,

(c) a point of inflection at $x = 3$, and a local maximum at $x = 0$,

(d) a point of inflection at $x = 3$, and a local minimum at $x = 0$,

(e) none of the above [3]

solution Differentiate once $y' = 5x^4 - 20x^3 = 0$ at $x = 0, 4$. Differentiate again, $y'' = 20x^3 - 60x^2$. So $y''(4) > 0$ and $x = 4$ is a minimum. Also $y''(3) = 0$, so $x = 3$ is a point of inflection. The first non-zero derivative at $x = 0$ is $y^{(4)} = 120 - 120x > 0$, so $x = 0$ is a maximum, hence (c).

6. If $\cos y = e^{-x}$, for $x \geq 0$, which of the following is $\dfrac{dy}{dx}$?

(a) $\cot y$, (b) $-\dfrac{e^{-x}}{\sqrt{1-e^{-2x}}}$, (c) $\tan y$, (d) $\dfrac{e^{-2x}}{\sqrt{1-e^{-x}}}$,

(e) none of the above [2]

solution: Differentiate implicitly:

$$-\sin y \frac{dy}{dx} = -e^{-x} \Rightarrow \frac{dy}{dx} = \frac{e^{-x}}{\sin y} = \cot y,$$

so (a)

7. If $y = \tan u$, where $u = v + \dfrac{1}{v}$ and $v = \ln x$, what is the value of $\dfrac{dy}{dx}$ at $x = e$?

(a) $\dfrac{1}{e}$, (b) -1 , (c) $\dfrac{2}{e^2}$, (d) $\sec^2(e)$,

(e) none of the above [3]

solution (e). the answer is zero, using chain rule.

8. If $n = 0, 1, 2, 3, \ldots$, for which values of n is it true that

$$\int_0^1 x^n \, dx = \int_0^1 (1-x)^n \, dx \, ?$$

(a) all of them , (b) even values , (c) odd values , (d) none of them ,

(e) none of the above [2]

solution: Integrating, we have $\dfrac{1}{n+1} = \dfrac{1}{n+1}$, so all given n, answer (a).

9. The integral $I = \displaystyle\int \frac{1}{1-\sin x} \, dx$ is equal to

(a) $\dfrac{2}{1-\tan(x/2)} + \mathrm{C}$, (b) $\cot x + \sec x + \mathrm{C}$, (c) $\dfrac{x}{1+\cos x} + \mathrm{C}$, (d) $\dfrac{2}{1+\cot(x/2)} + \mathrm{C}$,

(e) none of the above [3]

Solution Use substitution $t = \tan(x/2)$, with $dx = \dfrac{2}{1+t^2} \, dt$ and $\sin x = \dfrac{2t}{1+t^2}$, from formula sheet to rewrite integral:

$$I = \int \frac{1}{1-\frac{2t}{1+t^2}} \frac{2}{1+t^2} \, dt = \int \frac{2}{1+t^2-2t} \, dt = \int \frac{2}{(t-1)^2} \, dt = -\frac{2}{t-1} + C =$$

so (a).

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}} \, .$$

Which of the following statements is true?

(a) S_2 diverges, and S_1 has infinite radius of convergence,

(b) S_2 converges, and S_1 has radius of convergence $R = 1$,

(c) S_2 has radius of convergence $R = 1$, and S_1 diverges for $|x| > 1$,

(d) S_2 diverges, and S_1 has interval of convergence $[-1, 1)$,

(e) none of the above [3]

solution (d)

11. The Maclaurin series for $f(x) = \sqrt{1-x}$ to order two, with Lagrange remainder term is given by

$$1 - \frac{1}{2}x - \frac{1}{8}x^2 + R_2 \, .$$

Which of the following are the correct Lagrange remainder, and inequality regarding the uncertainty expressed in c?

(a) $R_2 = -\frac{x^3}{16(1-c)^{5/2}}$ and $0 < |c| < |x|$,

(b) $R_2 = -\frac{x^2}{8(1-c)^{3/2}}$ and $0 < c < x$,

(c) $R_2 = \left| \frac{x^3}{16(1-c)^{5/2}} \right|$ and $0 < |c| < |x|$,

(d $R_2 = -\frac{x^3}{16(1-x)^{5/2}}$ and $0 < c < x$,

(e) none of the above [3]

solution

$$f(x) = (1-x)^{1/2} \Rightarrow f'(x) = -\frac{1}{2}(1-x)^{-1/2} \Rightarrow f''(x) = -\frac{1}{4}(1-x)^{-3/2} \Rightarrow f'''(x) = -\frac{3}{8}(1-x)^{-5/2}$$

so the remainder is $R_2 = -\frac{x^3}{16(1-c)^{5/2}}$ where $0 < |c| < |x|$, so (a)

12. The first order ordinary differential equation (ODE)

$$xy' = \frac{1}{x}(y-1)$$

has general solution

(a) $y = Ce^{1/x} - 1$, (b) $y = 1 + Ce^{-1/x}$, (c) $y = 1 - e^{-1/x} + \mathrm{C}$, (d) $y = \dfrac{1}{x + Ce^{-1/x}}$,

(e) none of the above [3]

solution: Rewrite as standard linear first-order

$$\frac{dy}{dx} - \frac{1}{x^2}y = -\frac{1}{x^2}$$

Integrating factor is $e^{\int -\frac{1}{x^2}\,dx} = e^{1/x}$ so we multiply by the IF and rewrite as

$$\frac{d}{dx}\left(ye^{1/x}\right) = -\frac{1}{x^2}e^{1/x} \Rightarrow ye^{1/x} = -\int \frac{1}{x^2}e^{1/x}\,dx = e^{1/x} + C \Rightarrow y = 1 + Ce^{-1/x}$$

so (b)

13. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1\,,$$

which of the following is the complementary function?

(a) $y_c = c_1 xe^{-x} + c_2 e^{2x}$, (b) $y_c = c_1 e^{-x} + c_2 e^{2x}$,

(c) $y_c = c_1 e^{-x} + c_2 e^{-2x}$, (d) $y_c = c_1 e^{-2x} + c_2 xe^{x}$,

(e) none of the above. [2]

solution Auxiliary equation: $\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda = 2, -1$, so (b).

14. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1\,,$$

which of the following is a particular integral?

(a) $y_p = -x^2 + x - \dfrac{3}{2}$, (b) $y_p = e^{-x} + e^{2x}$, (c) $y_p = x^2 - x - \dfrac{1}{2}$, (d) $y_p = -x^2 - \dfrac{1}{2}$,

(e) none of the above. [2]

solution Try $y_p = ax^2 + bx + c \Rightarrow y_p' = 2ax + b \Rightarrow y_p'' = 2a$ and substitute into the ODE

$$2a - (2ax + b) - 2(ax^2 + bx + c) = 2x^2 + 2x - 1$$

equating coefficients we get $a = -1, b = 0$ and $c = -1/2$, so (d)

15. Given the linear second-order ODE

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 2x^2 + 2x - 1\,,$$

which of the following is the solution satisfying $y = 1$ and $y' = 0$ when $x = 0$?

(a) $y = -xe^{-x} - \frac{1}{2}e^{2x} - x^2 + x - \frac{3}{2}$, (b) $y = \frac{1}{2}e^{-x} - e^{-2x} + x^2 - x - \frac{1}{2}$,

(c) $y = e^{-x} + \frac{1}{2}e^{2x} - x^2 - \frac{1}{2}$, (d) $y = -\frac{1}{2}e^{-2x} + xe^{x} + e^{-x} + e^{2x}$,

(e) none of the above. [2]

solution With general solution $y = c_1e^{-x} + c_2e^{2x} - x^2 - \frac{1}{2}$, we implement $y(0) = 1$ and get

$$1 = c_1 + c_2 - \frac{1}{2}$$

Differentiate once, and implement $y'(0) = 0$:

$$y' = -c_1e^{-x} + 2c_2e^{2x} - 2x \Rightarrow 0 = -c_1 + 2c_2$$

Solve to get $c_1 = 1, c_2 = 1/2$, so (c).

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. a) For $z = x + iy$, use the exponential definition of the hyperbolic cosine function to obtain $\cosh(iz)$ in the form $A + Bi$, where A, B are real trigonometric and hyperbolic functions.

Hence, or otherwise, show that [7]

$$|\cosh(iz)|^2 = \frac{1}{2}\cosh 2y + \frac{1}{2}\cos 2x.$$

Solution: Begin with

$$\cosh(iz) = \cosh(ix - y) = \frac{e^{ix-y} + e^{-(ix-y)}}{2} = \frac{e^{ix}e^{-y} + e^{-ix}e^{y}}{2} = \frac{(\cos x + i\sin x)e^{-y} + (\cos x - i\sin x)e^{y}}{2}$$

having used Euler's formula for $e^{\pm ix}$. Hence,

$$\cosh(iz) = \cos x\frac{e^{y} + e^{-y}}{2} - i\sin x\frac{e^{y} - e^{-y}}{2} = \cos x\cosh y - i\sin x\sinh y,$$

as required. Furthermore,

$$|\cosh(iz)|^2 = \cos^2 x\cosh^2 y + \sin^2 x\sinh^2 y = \cos^2 x\cosh^2 y + (1 - \cos^2 x)(\cosh^2 y - 1) = \cosh^2 y + \cos^2 x - 1$$

$$= \frac{1}{2}(\cosh 2y + 1) + \frac{1}{2}(\cos 2x + 1) - 1 = \frac{1}{2}\cosh 2y + \frac{1}{2}\cos 2x, \text{ as required.}$$

b) Given the function

$$f(x) = \frac{1}{1 + e^{-x}},$$

obtain $f_e(x)$, the even part of $f(x)$. With no further calculation, write down the odd part, $f_o(x)$, and explain your reasoning. Hence, use these to sketch $f(x)$. [6]

Solution: Begin with the definition of even part:

$$f_e(x) = \frac{1}{2}[f(x) + f(-x)] = \frac{1}{2}\left(\frac{1}{1 + e^{-x}} + \frac{1}{1 + e^{x}}\right)$$

now multiply numerator/denominator of the second fraction by e^{-x}:

$$= \frac{1}{2}\left(\frac{1}{1 + e^{-x}} + \frac{e^{-x}}{1 + e^{-x}}\right) = \frac{1}{2}\left(\frac{1 + e^{-x}}{1 + e^{-x}}\right) = \frac{1}{2}.$$

Hence we use the fact that $f_o + f_e = f(x)$ to write

$$f_o(x) = f(x) - f_e(x) = \frac{1}{1 + e^{-x}} - \frac{1}{2}.$$

Sketch at end.

c) If y is a function of x, prove the inverse function rule $\dfrac{dx}{dy} = \dfrac{1}{\frac{dy}{dx}}$,

and obtain the corresponding expression for $\dfrac{d^2x}{dy^2}$ in terms of derivatives of y with respect to x. [7]

Solution:
First, let f and g be inverse functions, so that $y = g(x) \Leftrightarrow x = f(y)$, and composing the two gives $f(g(x)) = x$. Now differentiate both sides:

$$1 = \frac{d}{dx}(x) = \frac{d}{dx}[f(g(x)] = \frac{df}{dg}\frac{dg}{dx},$$

using the chain rule.

Now, as $y = g(x)$, we can equate $\dfrac{dy}{dx} = \dfrac{dg}{dx}$.

Similarly, as $x = f(y)$, we can equate $\dfrac{dx}{dy} = \dfrac{df}{dy} = \dfrac{df}{dg}$.

Hence, $1 = \dfrac{dy}{dx}\dfrac{dx}{dy}$, and the result follows.

Now differentiate both sides w.r.t. y:

$$\frac{d^2x}{dy^2} = \frac{d}{dy}\left(\frac{dx}{dy}\right) = \frac{d}{dy}\left(\frac{1}{\frac{dy}{dx}}\right) \qquad (*)$$

Where we have d/dy of a function of x, use the implicit chain rule:

$$\frac{d}{dy}\left[\left(\frac{dy}{dx}\right)^{-1}\right] = \frac{d}{dx}\left[\left(\frac{dy}{dx}\right)^{-1}\right]\frac{dx}{dy} = \left[-\left(\frac{dy}{dx}\right)^{-2}\frac{d}{dx}\left(\frac{dy}{dx}\right)\right]\frac{dx}{dy} = -\frac{d^2y}{dx^2}\left(\frac{dy}{dx}\right)^{-3},$$

where the inverse function rule is used in the last step.

2. a) Given the limit:

$$L = \lim_{x\to\infty} x^q \left[(2x+1)^{1/2} - (2x-1)^{1/2}\right]$$

find the value q so that L exists, and is non-zero. Also calculate L. [6]

Solution: Rewrite to enable binomial expansion:

$$(2x\pm 1)^{1/2} = (2x)^{1/2}\left(1\pm\frac{1}{2x}\right)^{1/2} = (2x)^{1/2}\left[1+\frac{1}{2}\left(\pm\frac{1}{2x}\right)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left(\pm\frac{1}{2x}\right)^2+\ldots\right]$$

Hence

$$x^q\left[(2x+1)^{1/2}-(2x-1)^{1/2}\right] = x^q(2x)^{1/2}\left[1+\frac{1}{4x}-\frac{1}{32x^2}+\cdots-\left(1-\frac{1}{4x}-\frac{1}{32x^2}+\cdots\right)\right]$$

and even powers cancel, leaving

$$= \sqrt{2}\,x^{q+1/2}\left(\frac{1}{2x}+\frac{k_1}{x^3}+\frac{k_2}{x^5}+\ldots\right) = \sqrt{2}\left(\frac{1}{2}x^{q-1/2}+k_1x^{q-5/2}+k_2x^{q-9/2}+\ldots\right)$$

for constants k_i. For a non-zero limit to exist we need the power in the first term to be zero, hence $q = 1/2$. In the limit $x\to\infty$, all other terms with increasing negative powers of x vanish, leaving $L = 1/\sqrt{2}$.

b) Given the function $f(x) = (x^2-x-2)e^{2x}$, obtain all stationary points and use the second derivative to classify them. Find all intercepts, points of inflection and asymptotes, and sketch the function, explaining your results. [6]

Solution: Differentiate

$$f'(x) = (2x-1)e^{2x}+2(x^2-x-2)e^{2x} = (2x^2-5)e^{2x} = 0 \Rightarrow x = \pm\sqrt{5/2}$$

Again

$$f''(x) = 4xe^{2x}+2(2x^2-5)e^{2x} = 2(2x^2+2x-5)e^{2x}$$

Given that the exponential term is always positive, the sign of $f''(\pm\sqrt{5/2})$ is given by the polynomial

$$2(5/2)\pm\sqrt{5/2}-5 = \pm\sqrt{5/2}$$

so $x = \sqrt{5/2}$ is a local minimum, and $x = -\sqrt{5/2}$ a local maximum. We can approximate them with $1 < |\sqrt{2.5}| < 2$.

Intercepts are easy: $(0,-2)$ and $(-1,0),(2,0)$. Values at stationary points are determined by the polynomial again:
$f(\sqrt{5/2}) = 2(5/2-\sqrt{5/2}-2)\times$exponential term, giving a large negative number.
$f(-\sqrt{5/2}) = 2(5/2+\sqrt{5/2}-2)\times$exponential term, giving a small positive number.

Points of inflection are at

$$f''(x) = 2(2x^2+2x-5)e^{2x} = 0 \Rightarrow x = -\frac{1}{2}\pm\frac{1}{2}\sqrt{11} \approx -2,1$$

but can also be inferred. One horizontal asymptote, as $x \to -\infty$, due to the exponential term. As $x \to \infty$, the exponential takes over and $y \to \infty$. Sketch at end.

c) Given the integral

$$I_n \;=\; \int \frac{1}{(1-x^2)^n}\, dx,$$

show that

$$I_n = I_{n-1} + \int \frac{x^2}{(1-x^2)^n}\, dx\,.$$

Now integrate

$$\int \frac{x^2}{(1-x^2)^n}\, dx$$

by parts, and find an integral recurrence relation for I_n in terms of I_{n-1} and a function of x. [8]

Solution: For the first part, write:

$$I_{n-1} + \int \frac{x^2}{(1-x^2)^n}\, dx = \int \frac{1}{(1-x^2)^{n-1}}\, dx + \int \frac{x^2}{(1-x^2)^n}\, dx = \int \frac{1}{(1-x^2)^{n-1}} \frac{1-x^2}{1-x^2} + \frac{x^2}{(1-x^2)^n}\, dx$$

$$= \int \frac{1-x^2}{(1-x^2)^n} + \frac{x^2}{(1-x^2)^n}\, dx \;=\; \int \frac{1}{(1-x^2)^n}\, dx \;=\; I_n\,,$$

as required. For the second integral, let $u = x \Rightarrow du = dx$ and

$$dv = \frac{x}{(1-x^2)^n} \Rightarrow v = -\frac{1}{2} \int \frac{-2x}{(1-x^2)^n}\, dx - = \; -\frac{1}{2(1-n)} \frac{1}{(1-x^2)^{n-1}}$$

so that

$$\int \frac{x^2}{(1-x^2)^n}\, dx = \frac{x}{2(n-1)(1-x^2)^{n-1}} + \frac{1}{2(1-n)} \int \frac{1}{(1-x^2)^{n-1}}\, dx = \frac{x}{2(n-1)(1-x^2)^{n-1}} + \frac{1}{2(1-n)} I_{n-1}$$

Hence, put everything together:

$$I_n = I_{n-1} + \frac{x}{2(n-1)(1-x^2)^{n-1}} + \frac{1}{2(1-n)} I_{n-1} \;=\; \frac{3-2n}{2-2n} I_{n-1} + \frac{x}{2(n-1)(1-x^2)^{n-1}}\,,$$

as required.

3. a) Calculate the integral

$$I = \int \frac{1}{1-\cos x-\sin x}\, dx\,.$$

[5]

Solution: Use the substitution $t = \tan(x/2)$ so

$$\sin x = \frac{2t}{1+t^2}\,, \quad \cos x = \frac{1-t^2}{1+t^2}\,, \quad dx = \frac{2}{1+t^2}dt$$

and the integral becomes

$$I = \int \frac{1}{\left(1-\frac{1-t^2}{1+t^2}-\frac{2t}{1+t^2}\right)}\frac{2}{1+t^2}\,dt = \int \frac{2}{1+t^2-(1-t^2)-2t}\,dt = \int \frac{1}{t(t-1)}\,dt = \int \frac{1}{t-1}-\frac{1}{t}\,dt$$

$$= \ln(t-1)-\ln t+C = = \ln\left(1-\frac{1}{t}\right)+C = \ln\left(1-\frac{1}{\tan(x/2)}\right)+C$$

b) Obtain the general solution of the first-order ODE: [6]

$$y'(x) = \frac{x^2}{y^2}+\frac{2y}{x}\,.$$

Solution The ODE is homogenous, let $v = y/x \Rightarrow \dfrac{dy}{dx} = x\dfrac{dv}{dx}+v$ and the ODE becomes

$$x\frac{dv}{dx}+v = \frac{1}{v^2}+2v \Rightarrow x\frac{dv}{dx} = \frac{1}{v^2}+v = \frac{1+v^3}{v^2}$$

which is separable, so

$$\int \frac{v^2}{1+v^3}\,dv = \int \frac{1}{x}\,dx \Rightarrow \frac{1}{3}\ln(1+v^3) = \ln x+c \Rightarrow [1+(y/x)^3]^{1/3} = Ax \Rightarrow 1+(y/x)^3 = Kx^3 \Rightarrow$$

$$y^3 = x^3(Kx^3-1) \Rightarrow y = x\left(Kx^3-1\right)^{1/3}$$

is the general solution.

c) We seek a series solution of the second-order linear ODE

$$(1+x^2)\frac{d^2y}{dx^2}+\frac{3}{2}x\frac{dy}{dx}-\frac{1}{2}y = 0$$

satisfying the initial conditions $y = 1$ and $y' = 0$, when $t = 0$.

(i) Use Leibnitz' theorem to differentiate the series n times. [3]

Solution: Differentiate using Leibnitz:

$$\frac{d^n}{dx^n}\left[(1+x^2)\frac{d^2y}{dx^2}\right] = (1+x^2)y^{(n+2)}(x)+n(2x)y^{(n+1)}(x)+\frac{n(n-1)}{2}(2)y^{(n)}(x)$$

and

$$\frac{d^n}{dx^n}\left[x\frac{dy}{dx}\right] \;=\; xy^{(n+1)}(x)+n(1)y^{(n)}(x)$$

Put together into the ODE to obtain

$$(1+x^2)y^{(n+2)}(x)+2nxy^{(n+1)}(x)+n(n-1)y^{(n)}(x)+\frac{3}{2}xy^{(n+1)}(x)+\frac{3}{2}ny^{(n)}(x)-\frac{1}{2}y^{(n)}(x)=0\,.$$

Simplification in the next part.

(ii) Let $x=0$ in your result from (i) to obtain a recurrence relation

$$y^{(n+2)}(0)=f(n)y^{(n)}(0)\;, \qquad (n\geq 0)$$

where $f(n)$ needs to be determined, and $y^{(k)}(0)$ is y differentiated k times, and evaluated at $x=0$. [3]

Solution: Letting $x=0$ above, we get

$$y^{(n+2)}(0)+2n(0)y^{(n+1)}(0)+n(n-1)y^{(n)}(0)+\frac{3}{2}(0)y^{(n+1)}(0)+\frac{3}{2}ny^{(n)}(0)-\frac{1}{2}y^{(n)}(0)=0\,,$$

and tidy up leaving

$$y^{(n+2)}(0) \;=\; \left(\frac{1}{2}-\frac{3}{2}n-n(n-1)\right)y^{(n)}(0) \;=\; \left(\frac{1}{2}-\frac{1}{2}n-n^2\right)y^{(n)}(0)$$

(iii) Use the recurrence relation and initial conditions to obtain the first three non-zero terms of the Maclaurin series of $y(x)$. [3]

Solution: We have the initial conditions $y(0)=1=y^{(0)}(0)$ (the function differentiated zero times is just the function) and $y'(0)=0=y^{(1)}(0)$, so now in the recurrence relation:

let $n=0,\ \Rightarrow\ y^{(2)}(0)=\frac{1}{2}y^{(0)}(0)=\frac{1}{2}y(0)=\frac{1}{2}.$

let $n=1,\ \Rightarrow\ y^{(3)}(0)=\left(\frac{1}{2}-\frac{1}{2}-1\right)y^{(1)}(0)=0$, as $y'(0)=0$.

let $n=2,\ \Rightarrow\ y^{(4}(0)=\left(\frac{1}{2}-1-4\right)y^{(2)}(0)=-\frac{9}{2}\times\frac{1}{2}=-\frac{9}{4}$ and we have enough terms for the series:

$y(x)=y(0)+y'(0)x+\dfrac{y''(0)}{2!}x^2+\cdots$ so the first three non-zero terms are

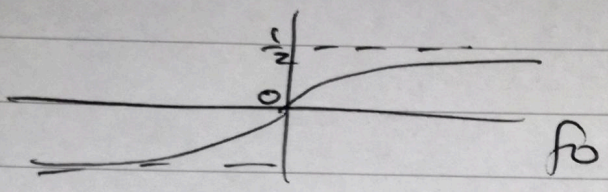
$$y(x) \;=\; 1+\frac{1/2}{2!}x^2+\frac{-9/4}{4!}x^4+\ldots=1+\frac{1}{4}x^2-\frac{3}{32}x^4+\ldots$$

Question B

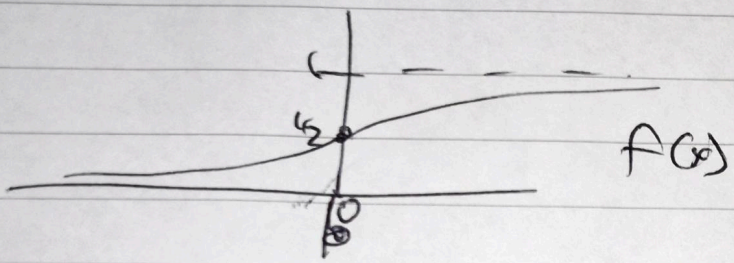
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16) f_0 is odd part and $f_0(0) = 0$.

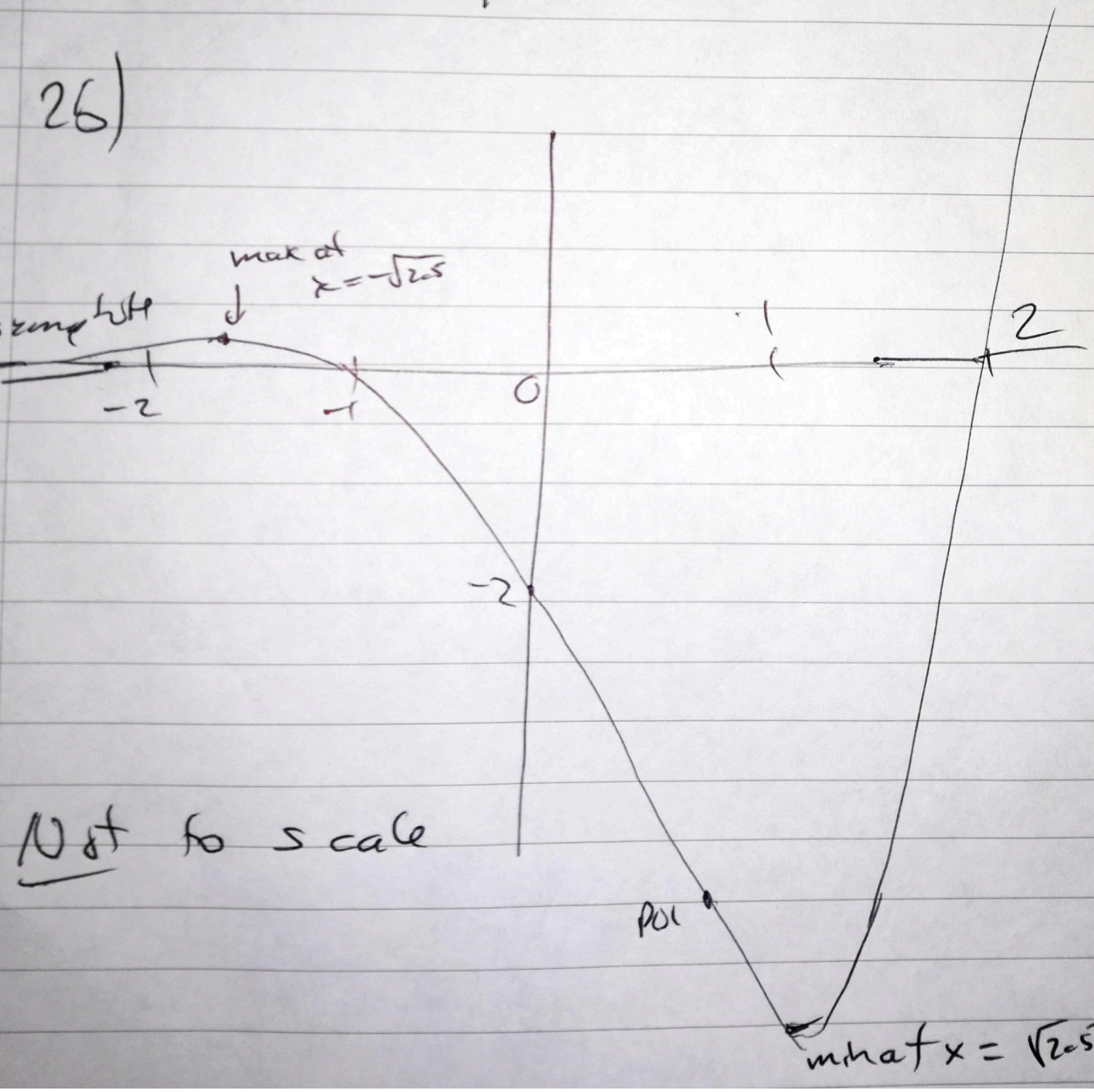
As $x \to \infty$, $e^{-x} \to 0$, $f_0 \to \frac{1}{2}$. By symmetry



so $f = f_0 + \frac{1}{2}$



26)



Not to scale